Lunga Owen Mtsweni

AI/ML-focused Software Engineer & Solutions Architect

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SUMMARY

Final-year Computer Science Diploma student (NQF 6) at Tshwane University of Technology, graduating April 2027. Sole architect of ALIS, a legal-tech compliance platform spanning Java Spring Boot, Python FastAPI/Celery, React/TypeScript and Cloudflare Workers. Strong in microservices, event-driven design and auth/security, with a career goal in AI/ML engineering. Currently on WIL placement at ICEP. Seeking a graduate programme or internship.

EXPERIENCE

Solutions Architect / Developer, ALIS (AI Legal Intelligence System)

Final-year flagship project | Sole architect and contributor

- Architected a platform that checks South African contracts (employment agreements, NDAs, leases) against BCEA, LRA, CPA and POPIA.
- Designed the multi-service system: Spring Boot orchestrator (AWS EC2) for auth, JWT and RBAC; FastAPI/Celery/Redis document pipeline (Docker, Azure VM); React/TypeScript frontend; Cloudflare Workers for edge APIs and a RAG chatbot.
- Built a seven-stage NLP pipeline: PDF extraction, NER, clause segmentation, Groq Llama 3.3 70B compliance reasoning, PDF report.
- Implemented Postgres full-text search (tsvector + GIN) on Supabase and a secure password-reset flow with Edge Functions and pgcrypto.
- Resolved production issues: LazyInitializationException-caused 403s, PgBouncer prepared-statement conflicts, Docker/Nginx misconfigurations.
- Leading Phase 3 (AI integration) with a 6+ member team.

WIL Placement, ICEP (Informatics Community Engagement Programme), TUT

Ongoing

- Gathered requirements and architected a fingerprint clocking system as microservices (attendance, auth, reporting); built the ASP.NET Core Web API that receives and stores clock-in/out events.
- Scoped an NFC/RFID wallet microservice for the Tshwane prepaid transit bus card backend: wallet and fare-deduction data model, Paystack top-up integration.
- Built a JavaScript dashboard showing real-time attendance records and daily summaries from the device APIs.

SELECTED PROJECTS

MarketLane | Multi-tenant marketplace for local and informal businesses (vendors, barbers, salons, restaurants) combining bookings with e-commerce. Microservices backend with per-service Postgres, RabbitMQ, Docker Compose and Nginx; Vite/React/TypeScript frontend.

Standalone Auth Microservices | ASP.NET Core + MySQL service on Clean Architecture with BCrypt and MailKit email verification; Python/FastAPI WebAuthn/FIDO2 biometric (face and fingerprint) service with JWT sessions.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript/JavaScript, C# (ASP.NET Core), SQL

Backend: Spring Boot, FastAPI, Celery, ASP.NET Core, REST API design, microservices

Frontend: React, TypeScript, Vite

Data & AI: PostgreSQL, Supabase, Redis, RabbitMQ, NLP pipelines, LLM integration (Groq/Llama), RAG, XGBoost, scipy

Infrastructure: AWS EC2, Azure VM, Cloudflare Workers/Pages, Docker, Docker Compose, Nginx, GitHub Actions

Security: JWT, RBAC, WebAuthn/FIDO2, BCrypt, OAuth flows

Practices: Solutions architecture, system design, Clean Architecture, event-driven design

EDUCATION & CERTIFICATIONS

Diploma in Computer Science (NQF 6) | Tshwane University of Technology, Soshanguve South Campus | Expected April 2027

Huawei ICT Academy: Overview of AI | In progress: HCIA-AI V4.0, AWS AI Practitioner (AIF-C01)

LINKS

GitHub: github.com/Lungaowen | LeetCode: leetcode.com/Joni-dev16